

MUHAMMAD SAAD FAREED

Senior Software Engineer

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Lahore, Pakistan

SUMMARY

Senior Backend Engineer with 3+ years of experience architecting high-throughput, distributed systems serving millions of transactions. Expert in Python (Django, FastAPI), microservices, and event-driven design (SQS, MQTT, gRPC). Delivered measurable impact: 80% reduction in API latency, 35% faster checkout, and 25% lower database load. Designed PCI DSS-compliant payment systems, multi-database architectures (PostgreSQL, Cassandra, Elasticsearch), and real-time data pipelines. Ranked among the Top 2% of Python developers globally on GitHub. Thrives in fast-paced, cross-functional engineering environments

EDUCATION

University of Engineering and Technology, Lahore

2018 – 2022

Bachelor of Science in Computer Science

WORK EXPERIENCE

Foodie | Software Engineer II

Mar 2025 – Present

- Owned the end-to-end backend lifecycle of a multi-tenant, high-throughput order platform; implemented PCI DSS-compliant checkout supporting 4 payment methods (card, cash, wallet, hybrid) using a distributed Saga pattern.
- Built a unified refund system that handles all cancellation scenarios (customer, vendor, system-initiated) across mixed-payment orders, ensuring full financial integrity and auditability at scale.
- Architected core revenue and payout infrastructure for vendors, delivery partners, and influencers; built influencer commission tracking via gRPC-based microservices across Python (Django, FastAPI) and Node.js.
- Designed a multi-database system: MongoDB for menus/social posts, Cassandra for high-write order events, and Elasticsearch for real-time search with sub-100ms response times.
- Built a real-time order notification backbone using MQTT + Redis pub/sub, delivering stage-specific updates to customers and store managers instantly.
- Led a monolith-to-microservices migration, extracting payments, promotions, and orders into independent, scalable services.

Agrilift.ai | Software Engineer I

May 2024 – Apr 2025

- Integrated AWS SQS for async data processing in germination and crop health pipelines, reducing database load by 25% and improving system resilience.
- Resolved critical API performance bottlenecks – reduced calculation time by 40% through profiling and query optimisation.
- Built a Root Cause Analysis (RCA) feature for field teams, cutting crop health issue identification time by 25% and improving agricultural decision accuracy.
- Developed weather data utilities (hourly conditions, anomaly detection) by analysing 5 years of historical data in collaboration with Data Science and Analytics teams.

CodeViz Technologies | Software Engineer

Jan 2023 – May 2024

- Built and maintained full-stack applications using Vue, NuxtJS, and Django REST Framework; improved frontend performance and integrated Google Analytics and D3.js visualisations.
- Wrote SQL migration scripts for MariaDB, optimised query performance, and designed scalable backend architectures for client-facing products.

TECHNICAL SKILLS

Languages & Frameworks: Python (Django, FastAPI, asyncio), Node.js, Go, TypeScript, SQL

Backend: REST/gRPC, SQS, MQTT, Microservices, Redis

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Elasticsearch, DynamoDB

DevOps: AWS (Lambda, SQS, EC2), Docker